

Music Generation I

AI in Music, Winter Term 2025/2026

22.01.2026

Exercise 1 [Markov Models, 4 points]

On Moodle you will find ten files containing abc notation of Irish Folk and Traditional French Dance pieces. Create **generative Markov models** for **each genre** with **different orders (1, 2, 4)**. Use these models to create pieces for every **genre-order combination (6 pieces in total)**, by choosing a **random valid starting** state and generating **16 bars of music**.

Exercise 2 [Evaluation, 2 points]

Create your own evaluation function that takes in a MIDI sequence (your generated pieces) and outputs a scalar score depending on quality criteria that you may freely choose.

Listen to the pieces you created (for example by using a website like <https://abc.rectanglered.com/>) and compare the scores your function outputs to fine-tune weights or other parameters you introduced. Evaluate the created pieces from Exercise 1 with regards to this function.

Exercise 3 [Evolutionary Algorithms, 4 points]

Write a simple Evolutionary Algorithm (EA) that creates a **random population of 10 pieces** from your **best performing Markov models (one from each genre)** and then optimises that population over **100 generations**. Implement the following simple mutations, one of which will be randomly applied to a child individual at every generation:

- Add a note with random pitch and value, at a randomly chosen index
- Remove a random note
- Change the pitch and/or value of a random note.

Run your algorithm **once for every Markov model you made** and use your **evaluation function from Exercise 2 as fitness function**. In every generation, create one child of each individual in a population of size 10 and apply one mutation. Keep the child individual if its fitness is better than its parent's fitness (a so-called (10 + 10) evolutionary strategy). Compare whether the EA is able to optimise your evaluation function and listen to the pieces it created.